SIH26189

AI-Powered Criminal Network Analysis System

Complete Project Architecture — Every Major Module

1. High-Level Architecture

```
INVESTIGATOR
↓
WEB DASHBOARD / UI
↓
API GATEWAY
↓
Auth/RBAC | Case Management | Data Ingestion | Investigation
↓
Document Processing
■■■ Text PDF → PyMuPDF
■■■ Scanned PDF → OCR
↓
NLP / Entity Extraction
↓
Entity Resolution
↓
Human Review
↓
Graph Builder
↓
NEO4J
↓
Network Analysis | Money Trail | Path Analysis
↓
Investigation UI → Graph | Timeline | Alerts | Reports
```

2. Authentication & User Management

Roles: Admin, Senior Investigator, Investigator, Analyst.
Features: login, session management, RBAC, case-level permissions, security controls and audit logging.

3. Case Management

Every investigation is organized inside a Case.
Case: ID, title, description, creator, assigned officers, status, priority, date, evidence and notes.
Features: create, assign, update, close/archive, evidence management, notes and case timeline.

4. Data Ingestion

Inputs:
• FIR: PDF / DOCX / TXT / scanned PDF
• CDR: CSV / Excel / JSON
• Bank: CSV / Excel / JSON
• Other legally available data: IP, vehicle, crypto-wallet and email records.
Pipeline: Upload → Validate → Parse → Normalize → Process.

5. Document Processing

PDF → type detection.
Text PDF → PyMuPDF/pdfplumber → text.
Scanned PDF → OCR (e.g. Tesseract) → text.
Then clean and normalize the text.

6. AI / NLP Entity Extraction

Example FIR: “Rahul Sharma used mobile 9876543210 and transferred Rs 5 lakh to account 1234567890.”
Output:
PERSON → Rahul Sharma
PHONE → 9876543210
AMOUNT → 500000 INR
BANK_ACCOUNT → 1234567890

Entities: Person, Phone, Bank Account, UPI ID, Email, IP, Vehicle, Crypto Wallet, Location, Organization, Date, Money, FIR.

7. Rule-Based Extraction

Use deterministic rules alongside NLP for obvious structured fields:
Phone numbers, IP addresses, bank-account patterns, amounts and dates.
Combined output = NLP + Regex/rules.

8. Entity Resolution & Deduplication

Different sources may contain: Rahul Sharma, Rahul S., R. Sharma, Rahul Kumar Sharma.
System suggests a possible same entity with confidence and supporting signals.
Matching signals: name similarity, phone/email, address/DOB where available, bank account, vehicle and known associations.
Investigator approves or rejects the merge.

9. Data Cleaning & Normalization

Standardize formats before graph creation.
Example: +91-9876543210 / 9876543210 → canonical phone.
■5,00,000 / Rs 500000 / 500000 INR → canonical monetary value.

10. Graph Construction

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Nodes:
Person, Phone, BankAccount, Transaction, IP, Vehicle, CryptoWallet, FIR, Location, Organization, Email.

Relationships:
Person USES Phone
Person OWNS BankAccount
Person DRIVES Vehicle
Person USES IP
Person OWNS CryptoWallet
Phone CALLED Phone
Account TRANSFERRED_TO Account
Person MENTIONED_IN FIR
Person LOCATED_AT Location
```

11. Graph Database Layer

Recommended: Neo4j.
Example:
Rahul → USES → Phone A → CALLED → Phone B → USED BY → Amit → OWNS → Bank A → TRANSFERRED → Bank B.

12. Investigation Search

Search by person, phone, bank account, vehicle, IP, crypto wallet or FIR.
Result: selected entity → direct links → related entities → paths → source evidence.

13. Multi-Hop Investigation

Allow 1–4 hop expansion.
Example: Rahul → Phone A → Phone B → Amit → Bank A.
Controls: 1 Hop | 2 Hop | 3 Hop | 4 Hop.

14. Money Trail Analysis

Account A → ■5L → Account B → ■3L → Account C → ■2L → Account D.
Show total transferred, transaction count, dates, transaction IDs, source/destination and full money-flow path.

15. Communication Network Analysis

CDR analysis: A → B → C → D.
Metrics: call count, duration, frequency, first/last contact and communication clusters.

16. Network / Graph Analytics

Degree Centrality → many connections.
Betweenness Centrality → structural bridges.
PageRank → structurally important nodes.
Community Detection → connected clusters.

Important: high centrality does NOT prove criminality. Use “high-centrality entity requiring further investigation.”

17. AI Investigation Assistant

AI assists investigators with graph queries and summaries.
Example: “Show connections between Rahul Sharma and Account 1234.”
AI returns the relevant path, hop count, entities and evidence references.
It should support—not replace—the investigator’s decision.

18. Anomaly / Pattern Detection

Flag patterns for review:
• Rapid fund movement
• Many accounts → one account
• One phone → many persons
• Sudden unusual transaction amount
• Unusual communication patterns
Use “Potential anomaly”, not “Confirmed criminal activity.”

19. Timeline Module

Chronological events:
01 Aug → FIR registered
05 Aug → Phone A calls Phone B
07 Aug → ■5L transferred
09 Aug → Vehicle event
12 Aug → Account B transfers ■3L
Support filtering and links back to evidence.

20. Geographic Analysis

Where legally available:
Person → Phone → Location.
Possible views: locations, movement history, common locations and transaction locations.

21. Evidence Management

Every extracted fact should retain provenance.
Example:
Rahul Sharma → FIR_1023.pdf → Page 4.
9876543210 → CDR_August.csv → Row 1832.
This allows graph facts to be traced back to source evidence.

22. Evidence Integrity

Evidence file → SHA-256 hash → stored integrity value.
If the file changes: old hash ≠ new hash → evidence integrity mismatch.

23. Audit Log

Record user, timestamp, action, case and relevant object/evidence.
Example: Officer A opened case → uploaded FIR → Officer B approved entity match → report generated.

24. Automated Investigation Report

PDF sections:
Case information
Investigator/date/status
Entity summary
Network summary
Major clusters
Money trails
Timeline
Evidence references
Investigation notes.

25. Frontend Architecture

```
React
■■■ Login
■■■ Dashboard
■■■ Cases
■ ■■■ Case List
■ ■■■ Case Details
■ ■■■ Evidence
■■■ Data Upload
■■■ Entity Review
■■■ Investigation
■ ■■■ Search
■ ■■■ Graph
■ ■■■ Timeline
■ ■■■ Money Trail
■ ■■■ Network Analysis
■■■ Alerts
■■■ Reports
■■■ Admin

Graph visualization: Cytoscape.js recommended.
```

26. Backend Architecture

```
React → Node.js/Express → REST API → services.
Services:
Auth, Case, Upload, Document, Entity, Graph, Investigation, Analytics, Report, Audit.
AI layer: Node.js ↔ FastAPI ↔ Python NLP/AI services.
```

27. AI Service Architecture

```
FastAPI
■■■ OCR
■■■ NER
■■■ Rule/Regex Extraction
■■■ Entity Resolution
■■■ Anomaly Detection
```

```
↓
Structured JSON
↓
Node.js API
↓
Neo4j / PostgreSQL
```

28. Database & Storage

Neo4j → graph relationships, paths, networks and transactions.
PostgreSQL → users, cases, permissions, metadata, audit logs and report metadata.
Object storage → FIR PDFs, images, CSVs, evidence and generated reports.

29. Complete End-to-End Data Flow

```
FIR / CDR / BANK
↓
DATA INGESTION
↓
FILE VALIDATION
↓
DOCUMENT PROCESSING
↓
TEXT / OCR
↓
ENTITY EXTRACTION
■■■ NER
■■■ REGEX
↓
ENTITY RESOLUTION
↓
HUMAN REVIEW
↓
GRAPH BUILDER
↓
NEO4J
↓
Network | Money Trail | Path Analysis
↓
INVESTIGATION UI
↓
Graph | Timeline | Alerts | Reports
```

30. SIH Demo Flow

```
LOGIN
↓
CREATE CASE
↓
UPLOAD FIR PDF
↓
PDF → TEXT
↓
ENTITY EXTRACTION
↓
SHOW ENTITIES
↓
INVESTIGATOR APPROVES
↓
UPLOAD CDR CSV
↓
UPLOAD BANK CSV
↓
ENTITY RESOLUTION
↓
BUILD NEO4J GRAPH
↓
SEARCH “Rahul Sharma”
↓
3-HOP GRAPH
↓
MONEY TRAIL
↓
CENTRALITY ANALYSIS
↓
TIMELINE
```

```
↓
GENERATE PDF REPORT
```

31. MVP Priority

Must-have:
1. Login/RBAC
2. Case management
3. PDF upload
4. PDF → text
5. Entity extraction
6. Entity review
7. CDR CSV upload
8. Bank CSV upload
9. Entity resolution
10. Neo4j graph
11. Graph visualization
12. 1–4 hop search
13. Money trail
14. Centrality
15. Timeline
16. Evidence provenance
17. PDF report
18. Audit log

Phase 2:
OCR, AI assistant, anomaly detection, community detection, geographic visualization, crypto analysis.

32. Core Project Principle

Heterogeneous Data
→ Entity Extraction
→ Entity Resolution
→ Knowledge Graph
→ Graph Analytics
→ Investigator Decision

The system organizes and connects evidence; it should not automatically declare a person criminal based on graph position, centrality or an AI-generated score.